Predicting Abnormal Lower Airway Disease using Cough Recordings

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Background

Motivation

Chest X-Ray (CXR) is a primary tool for identifying lower airway disease abnormalities (including TB), but access is limited in resource-constrained settings. Cough acoustics may capture airway pathology and provide a non-invasive, low-cost, alternative smartphone-based screening tool.

Objective

to evaluate the accuracy of cough audio features to classify CXR abnormality status in adults undergoing TB evaluation across 5 countries.

Data

Population & setting

Adults undergoing TB evaluation at outpatient clinics in Uganda (n=300), Philippines (n=248), Vietnam (n=241), South Africa (n=184), India (n=85)

Total: 1,058 recordings (~10 sec each)

Platform: Swaasa app developed by Salcit Technologies

Target Outcome: Composite CXR abnormality status based on independent radiologist annotation + AI-based abnormality score

Methods

We evaluated three cough-based classification pipelines to predict *chest X-ray (CXR) abnormality* status in adults undergoing tuberculosis evaluation across five countries.

1. **Full Recording Acoustic Pipeline:** Acoustic features extracted from entire ~10-second cough recordings.
2. **Segmented Acoustic Pipeline:** Recordings segmented into individual cough events using energy-based detection. Features extracted per event and aggregated at the subject level using weighted probabilities.
3. **Pretrained Embedding Pipeline (Google HeAR):** Individual cough events transformed into pretrained audio embeddings (512D) and compression. Subject-level predictions aggregated across events using mean probabilities.

Acoustic Features: MFCCs, mel spectrogram statistics, spectral features, temporal features, STFT measures, and voice quality metrics.

Feature Selection: Univariate Logistic Regression ($p<0.4$)

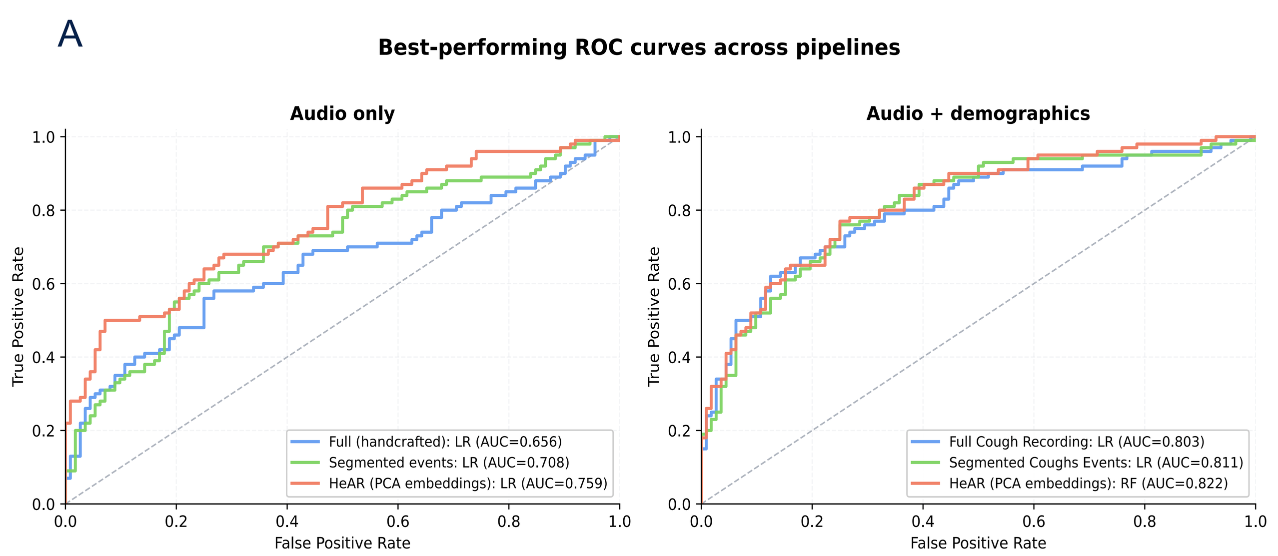
Baseline Models: Logistic Regression, SVM, and Random Forest (80-20 train-test split).

Performance evaluation: AUROC, sensitivity, specificity, accuracy, and subgroup analyses.

Pipeline and Results

Figure 1. Best Performing ROC Curves across Pipelines

ROC curves for the best-performing classifier per pipeline, shown without (A) and with (B) clinical demographics. HeAR (PCA) achieves the strongest audio-only performance (LR, AUROC 0.759); all pipelines converge to AUROC 0.80–0.82 with demographics.



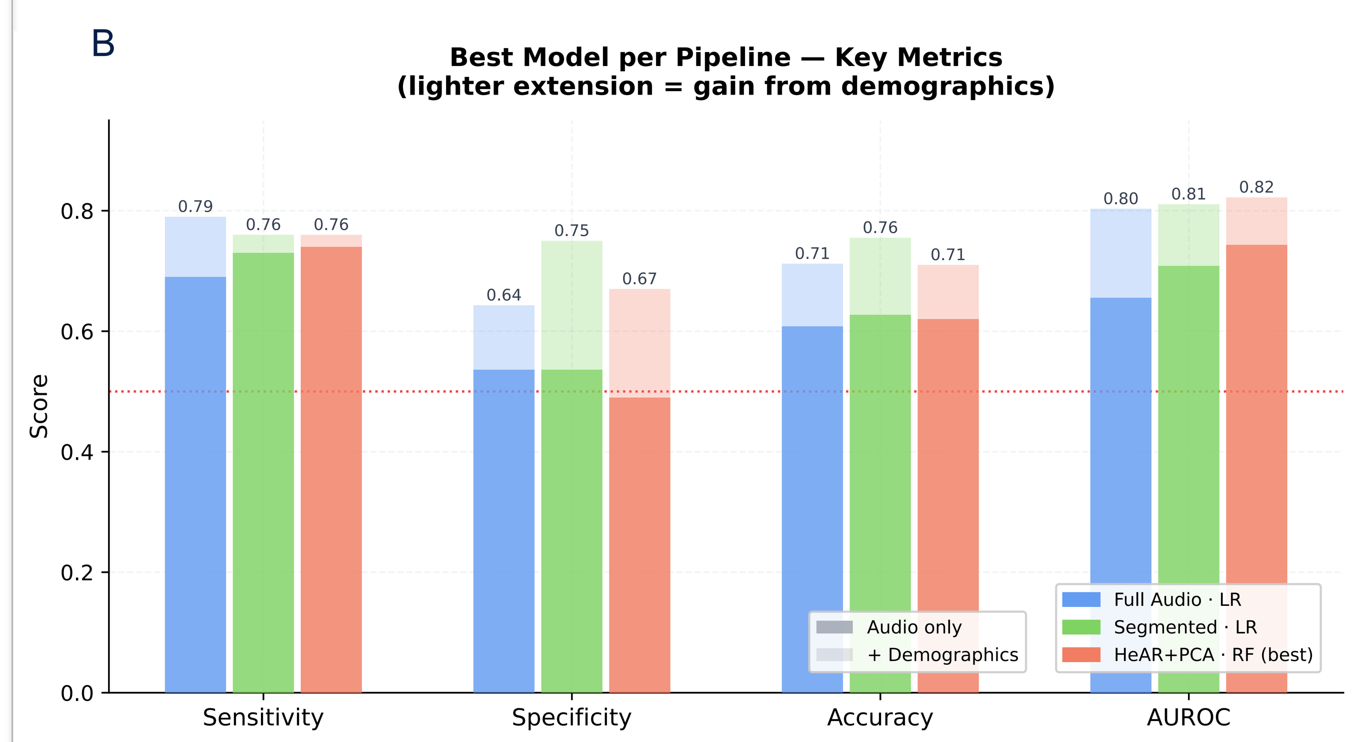


Figure 2. Subgroup Performance Breakdown

We break down the demographic features from the best performing model using audio features only (HeAR PCA embeddings with Logistic Regression). We observe heterogeneity across country effects and age groups.

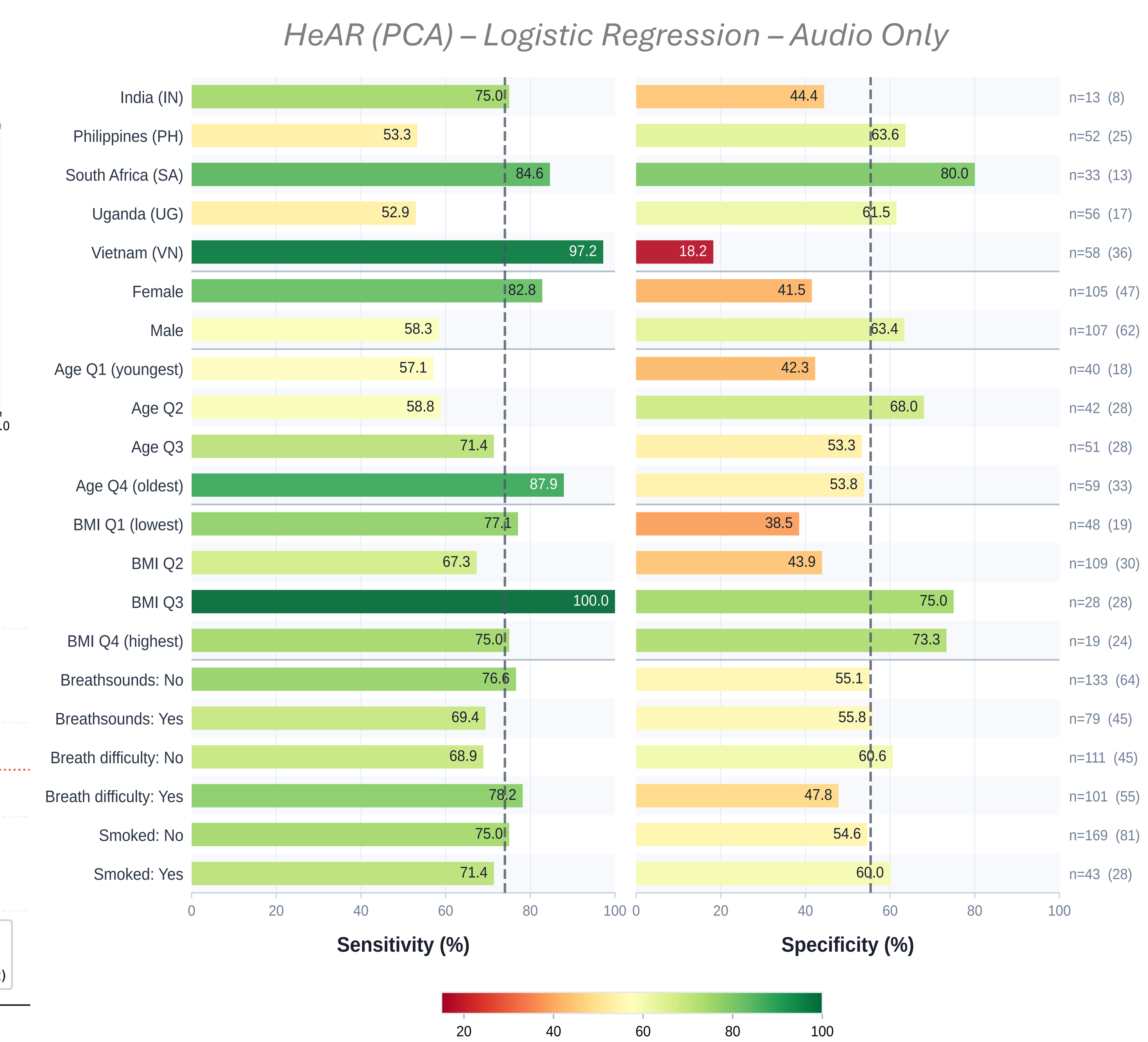


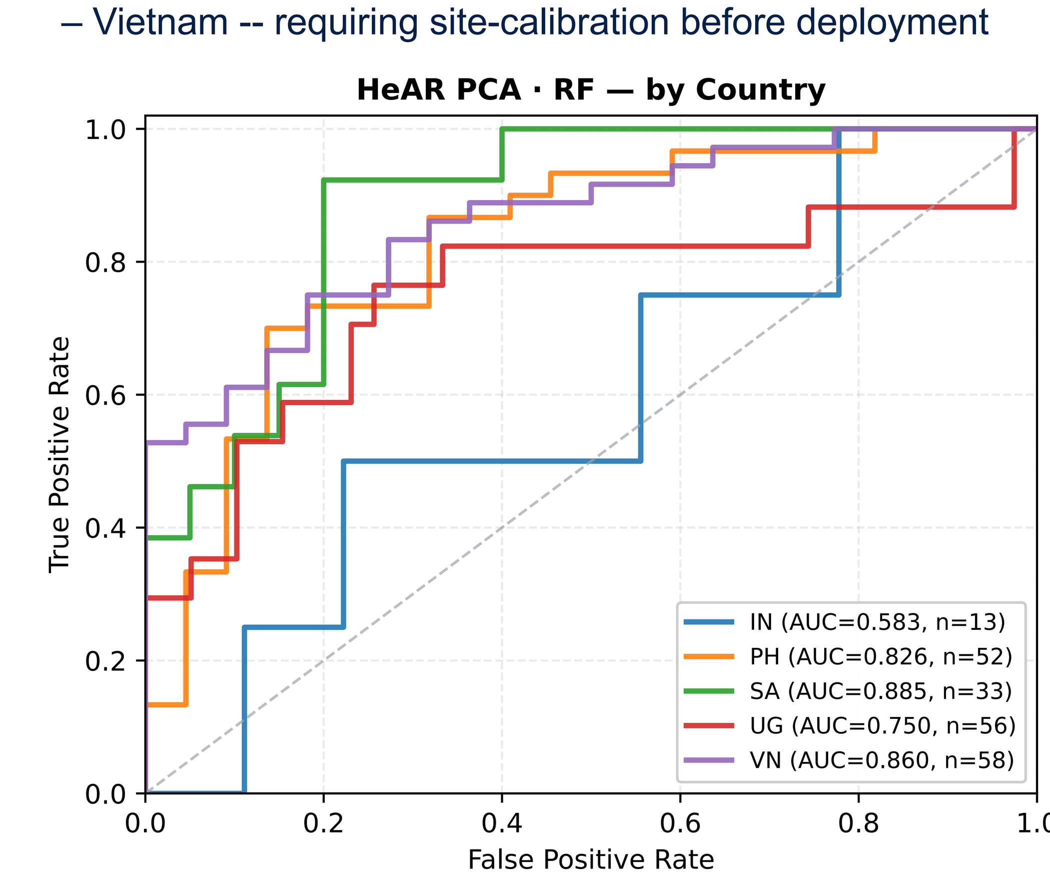
Table. Model performance (AUROC) across three pipelines, classifiers, and feature sets. Validation set (n=212).

Pipeline / Feature Set	RF	SVM	LR
Audio Features Only (No Demographics)			
Pipeline A – Full recording	0.628	0.637	**0.656**
Pipeline B – Segmented events	0.653	0.664	**0.708**
Pipeline C – HeAR (PCA)	0.741	0.753	**0.759†**
Pipeline C – HeAR (AE128)	0.721	0.739	**0.748**
Audio Features (w/ Clinical Demographics)			
Pipeline A – Full recording	0.761	0.758	**0.803**
Pipeline B – Segmented events	0.794	0.788	**0.811**
Pipeline C – HeAR (PCA)	**0.822†**	0.817	0.810
Pipeline C – HeAR (AE128)	**0.822**	0.812	0.812

† Best performing model for Audio Features Only vs. w/ Demographics/ RF = Random Forest; SVM = Support Vector Machine; LR = Logistic Regression; AE = Autoencoder; PCA = Principal Component Analysis.

Figure 3. Subgroup ROC Curve (Country)

We observe model performance is unstable across countries – Vietnam -- requiring site-calibration before deployment



Findings

Metric Highlights

- **Best audio-only model**: HeAR (PCA) LR [AUROC 0.759, sensitivity 74%, specificity 55%]
- **Best overall model**: HeAR (PCA) RF + demographics [AUROC 0.822, sensitivity 76%, specificity 67%]
- +0.08–0.14 AUROC lift from demographics across all three pipelines
- Segmented LR + demographics: comparable performance with AUROC 0.81
- Sensitivity increased with age: Q1 57% to Q4 88%
- Vietnam: sensitivity 97.2%, specificity 18.2% -- highest country-level imbalance across all models

Discussion + Next Steps

- Cough audio alone achieves moderate accuracy (AUROC ~0.76), supporting potential as a low-cost, non-invasive screening tool.
- Pretrained HeAR embeddings outperform handcrafted acoustic features without any TB-specific training, suggesting transfer learning offers a scalable path forward for resource-limited settings
- Clinical demographics provide the largest performance lift (+0.08–0.14 AUROC); whether cough adds independent signal beyond demographics alone remains to be established.
- Substantial country-level heterogeneity (sensitivity 53–97%) indicates a single global threshold will not perform equitably; site-specific calibration is needed before deployment.
- Best combined models (AUROC 0.81–0.82) approach but do not yet meet WHO TPP requirements (≥90% sensitivity, ≥80% specificity).

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